Terminal Output From Client Side

1. All Topics Request

```
========================================
SINGLE-THREADED EXPERIMENT: 1 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3726
[1]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=32.404993 MBps=1.873

Clients: 1
Total bytes: 60686716
Batch wall time: 32.434836 seconds
Aggregate throughput: 1.871035 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 2 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3735
Starting client 2 requesting topic: cavallo
[2] 3736
[1]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=43.430785 MBps=1.397

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=26.990542 MBps=1.386

Clients: 2
Total bytes: 98107225
Batch wall time: 54.924826 seconds
Aggregate throughput: 1.786209 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 4 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3747
Starting client 2 requesting topic: cavallo
[2] 3748
Starting client 3 requesting topic: elefante
[3] 3749
Starting client 4 requesting topic: farfalla
[4] 3750
[1]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[3]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
```

2. Single Thread


```
========================================
SINGLE-THREADED EXPERIMENT: 1 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3726
[1]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=32.404993 MBps=1.873

Clients: 1
Total bytes: 60686716
Batch wall time: 32.434836 seconds
Aggregate throughput: 1.871035 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 2 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3735
Starting client 2 requesting topic: cavallo
[2] 3736
[1]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=43.430785 MBps=1.397

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=26.990542 MBps=1.386

Clients: 2
Total bytes: 98107225
Batch wall time: 54.924826 seconds
Aggregate throughput: 1.786209 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 4 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3747
Starting client 2 requesting topic: cavallo
[2] 3748
Starting client 3 requesting topic: elefante
[3] 3749
Starting client 4 requesting topic: farfalla
[4] 3750
[1]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[3]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
```

```
--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=41.162247 MBps=1.474

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=23.828173 MBps=1.570

--- Client 3: elefante ---
[client] topic=elefante expected_files=1446
METRIC mode=part1 status=OK files=1446 bytes=58509473 seconds=150.141883 MBps=0.390

--- Client 4: farfalla ---
[client] topic=farfalla expected_files=2112
METRIC mode=part1 status=OK files=2112 bytes=61877367 seconds=48.543904 MBps=1.275

Clients: 4
Total bytes: 218494065
Batch wall time: 229.010527 seconds
Aggregate throughput: 0.954079 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 6 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3781
Starting client 2 requesting topic: cavallo
[2] 3782
Starting client 3 requesting topic: elefante
[3] 3783
Starting client 4 requesting topic: farfalla
[4] 3784
Starting client 5 requesting topic: gallina
[5] 3785
Starting client 6 requesting topic: gatto
[6] 3786
[1]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[3]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[5]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[6]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=36.617794 MBps=1.657

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=28.720538 MBps=1.303

--- Client 3: elefante ---
[client] topic=elefante expected_files=1446
METRIC mode=part1 status=OK files=1446 bytes=58509473 seconds=39.364946 MBps=1.486
```

```
--- Client 4: farfalla ---
[client] topic=farfalla expected_files=2112
METRIC mode=part1 status=OK files=2112 bytes=61877367 seconds=39.265665 MBps=1.576

--- Client 5: gallina ---
[client] topic=gallina expected_files=3098
METRIC mode=part1 status=OK files=3098 bytes=51842424 seconds=28.592147 MBps=1.813

--- Client 6: gatto ---
[client] topic=gatto expected_files=1668
METRIC mode=part1 status=OK files=1668 bytes=151144282 seconds=103.446993 MBps=1.461

Clients: 6
Total bytes: 421480771
Batch wall time: 257.835570 seconds
Aggregate throughput: 1.634688 MB/s

========================================
SINGLE-THREADED EXPERIMENT: 8 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 3813
Starting client 2 requesting topic: cavallo
[2] 3814
Starting client 3 requesting topic: elefante
[3] 3815
Starting client 4 requesting topic: farfalla
[4] 3816
Starting client 5 requesting topic: gallina
[5] 3817
Starting client 6 requesting topic: gatto
[6] 3818
Starting client 7 requesting topic: mucca
[7] 3819
Starting client 8 requesting topic: pecora
[8] 3820
[1]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[3]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[5]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[6]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[7]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1
[8]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/single_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=54.535297 MBps=1.113

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=23.962580 MBps=1.562

--- Client 3: elefante ---
[client] topic=elefante expected_files=1446
```

```
--- Client 4: farfalla ---
[client] topic=farfalla expected_files=2112
METRIC mode=part1 status=OK files=2112 bytes=61877367 seconds=35.195725 MBps=1.758

--- Client 5: gallina ---
[client] topic=gallina expected_files=3098
METRIC mode=part1 status=OK files=3098 bytes=51842424 seconds=44.257444 MBps=1.171

--- Client 6: gatto ---
[client] topic=gatto expected_files=1668
METRIC mode=part1 status=OK files=1668 bytes=151144282 seconds=82.626206 MBps=1.829

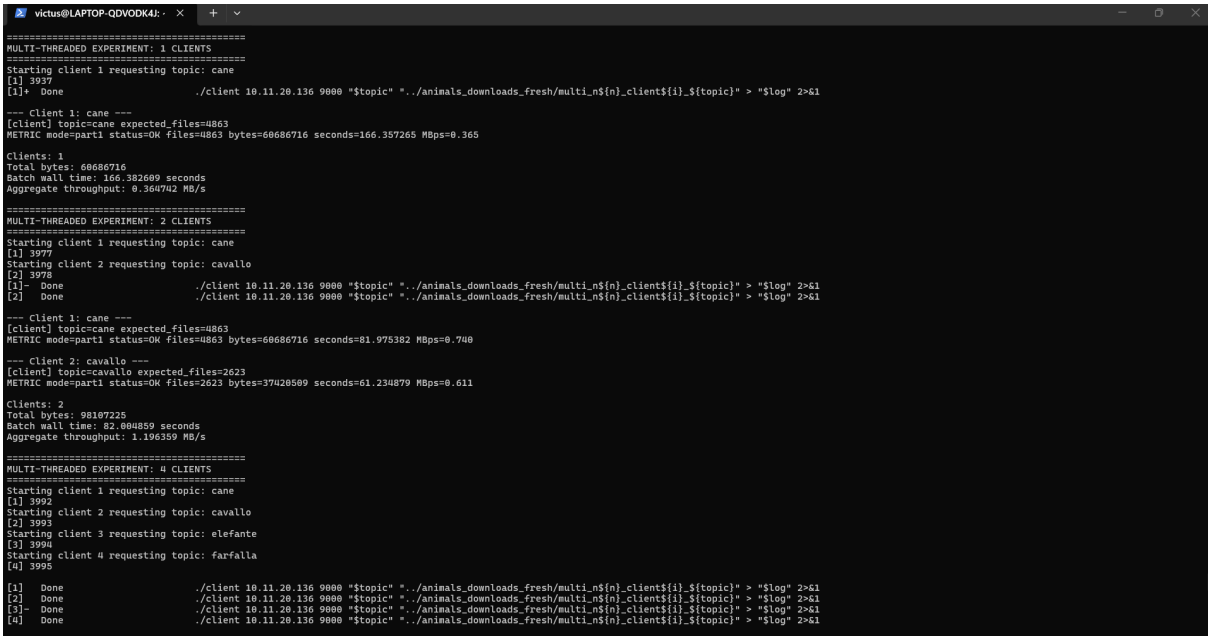
--- Client 7: mucca ---
[client] topic=mucca expected_files=1866
METRIC mode=part1 status=OK files=1866 bytes=25833171 seconds=23.844026 MBps=1.083

--- Client 8: pecora ---
[client] topic=pecora expected_files=1820
METRIC mode=part1 status=OK files=1820 bytes=62629703 seconds=44.789878 MBps=1.398

Clients: 8
Total bytes: 509943645
Batch wall time: 304.707492 seconds
Aggregate throughput: 1.673551 MB/s

========== FINAL CSV ==========
client_count,record_type,client_id,topic_name,transfer_status,files_received,total_bytes_transferred,client_transfer_time_seconds,client_transfer_rate_MBps,batch_wall_time_seconds,aggregate_throughput_MBps
1,client,1,cane,OK,4863,60686716,32.404993,1.873,32.434836,0
1,aggregate,0,ALL_TOPICS,OK,0,60686716,0,0,32.434836,1.871035
2,client,1,cane,OK,4863,60686716,43.430785,1.397,54.924826,0
2,client,2,cavallo,OK,2623,37420509,26.990542,1.386,54.924826,0
2,aggregate,0,ALL_TOPICS,OK,0,98107225,0,0,54.924826,1.786209
4,client,1,cane,OK,4863,60686716,41.162247,1.474,229.010527,0
4,client,2,cavallo,OK,2623,37420509,23.828173,1.570,229.010527,0
4,client,3,elefante,OK,1446,58509473,150.141883,0.390,229.010527,0
4,client,4,farfalla,OK,2112,61877367,48.843984,1.275,229.010527,0
4,aggregate,0,ALL_TOPICS,OK,0,218494065,0,0,229.010527,0.954079
6,client,1,cane,OK,4863,60686716,36.617794,1.657,257.835570,0
6,client,2,cavallo,OK,2623,37420509,28.720538,1.303,257.835570,0
6,client,3,elefante,OK,1446,58509473,39.364946,1.486,257.835570,0
6,client,4,farfalla,OK,2112,61877367,39.265665,1.576,257.835570,0
6,client,5,gallina,OK,3098,51842424,28.592147,1.813,257.835570,0
6,client,6,gatto,OK,1668,151144282,103.446993,1.461,257.835570,0
6,aggregate,0,ALL_TOPICS,OK,0,421080771,0,0,257.835570,1.634688
8,client,1,cane,OK,4863,60686716,54.535297,1.113,304.707492,0
8,client,2,cavallo,OK,2623,37420509,23.962580,1.562,304.707492,0
8,client,3,elefante,OK,1446,58509473,38.745143,1.510,304.707492,0
8,client,4,farfalla,OK,2112,61877367,35.195725,1.758,304.707492,0
8,client,5,gallina,OK,3098,51842424,44.257444,1.171,304.707492,0
8,client,6,gatto,OK,1668,151144282,82.626206,1.829,304.707492,0
8,client,7,mucca,OK,1866,25833171,23.844026,1.083,304.707492,0
8,client,8,pecora,OK,1820,62629703,44.789878,1.398,304.707492,0
8,aggregate,0,ALL_TOPICS,OK,0,509943645,0,0,304.707492,1.673551
```

3. Multi Thread



```
--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=113.652754 MBps=0.534

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=81.742508 MBps=0.458

--- Client 3: elefante ---
[client] topic=elefante expected_files=1446
METRIC mode=part1 status=OK files=1446 bytes=58509473 seconds=89.494624 MBps=0.654

--- Client 4: farfalla ---
[client] topic=farfalla expected_files=2112
METRIC mode=part1 status=OK files=2112 bytes=61877367 seconds=102.811036 MBps=0.602

Clients: 4
Total bytes: 218494065
Batch wall time: 113.687690 seconds
Aggregate throughput: 1.921880 MB/s

========================================
MULTI-THREADED EXPERIMENT: 6 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 4021
Starting client 2 requesting topic: cavallo
[2] 4022
Starting client 3 requesting topic: elefante
[3] 4023
Starting client 4 requesting topic: farfalla
[4] 4024
Starting client 5 requesting topic: gallina
[5] 4025
Starting client 6 requesting topic: gatto
[6] 4026
[1]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[2]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[3]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[5]-  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[6]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1

--- Client 1: cane ---
[client] topic=cane expected_files=4863
METRIC mode=part1 status=OK files=4863 bytes=60686716 seconds=216.444387 MBps=0.280

--- Client 2: cavallo ---
[client] topic=cavallo expected_files=2623
METRIC mode=part1 status=OK files=2623 bytes=37420509 seconds=85.223242 MBps=0.439

--- Client 3: elefante ---
[client] topic=elefante expected_files=1446
METRIC mode=part1 status=OK files=1446 bytes=58509473 seconds=448.453088 MBps=0.130
```

```
--- Client 4: farfalla ---
[client] topic=farfalla expected_files=2112
METRIC mode=part1 status=OK files=2112 bytes=61877367 seconds=181.440421 MBps=0.341

--- Client 5: gallina ---
[client] topic=gallina expected_files=3098
METRIC mode=part1 status=OK files=3098 bytes=51842424 seconds=196.813469 MBps=0.263

--- Client 6: gatto ---
[client] topic=gatto expected_files=1668
METRIC mode=part1 status=OK files=1668 bytes=151144282 seconds=494.491856 MBps=0.306

Clients: 6
Total bytes: 421480771
Batch wall time: 494.636647 seconds
Aggregate throughput: 0.852102 MB/s

========================================
MULTI-THREADED EXPERIMENT: 8 CLIENTS
========================================
Starting client 1 requesting topic: cane
[1] 4081
Starting client 2 requesting topic: cavallo
[2] 4082
Starting client 3 requesting topic: elefante
[3] 4083
Starting client 4 requesting topic: farfalla
[4] 4084
Starting client 5 requesting topic: gallina
[5] 4085
Starting client 6 requesting topic: gatto
[6] 4086
Starting client 7 requesting topic: mucca
[7] 4087
Starting client 8 requesting topic: pecora
[8] 4088


^C
[4]   Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1
[8]+  Done                    ./client 10.11.20.136 9000 "$topic" "../animals_downloads_fresh/multi_n${n}_client${i}_${topic}" > "$log" 2>&1

========== FINAL MULTI-THREADED CSV ==========
client_count,record_type,client_id,topic_name,transfer_status,files_received,total_bytes_transferred,client_transfer_time_seconds,client_transfer_rate_MBps,batch_wall_time_seconds,aggregate_throughput_MBps
1,client,1,cane,OK,4863,60686716,166.357265,0.365,166.382609,0.0
1,aggregate,0,ALL_TOPICS,OK,0,60686716,0.0,0.0,166.382609,0.364742
2,client,1,cane,OK,4863,60686716,81.975382,0.74,82.004859,0.0
2,client,2,cavallo,OK,2623,37420509,61.234879,0.611,82.004859,0.0
2,aggregate,0,ALL_TOPICS,OK,0,98107225,0.0,0.0,82.004859,1.196359
4,client,1,cane,OK,4863,60686716,113.652754,0.534,113.68769,0.0
4,client,2,cavallo,OK,2623,37420509,81.742508,0.458,113.68769,0.0
4,client,3,elefante,OK,1446,58509473,89.494624,0.654,113.68769,0.0
4,client,4,farfalla,OK,2112,61877367,102.811036,0.602,113.68769,0.0
4,aggregate,0,ALL_TOPICS,OK,0,218494065,0.0,0.0,113.68769,1.92188
6,client,1,cane,OK,4863,60686716,216.444387,0.28,494.636647,0.0
```

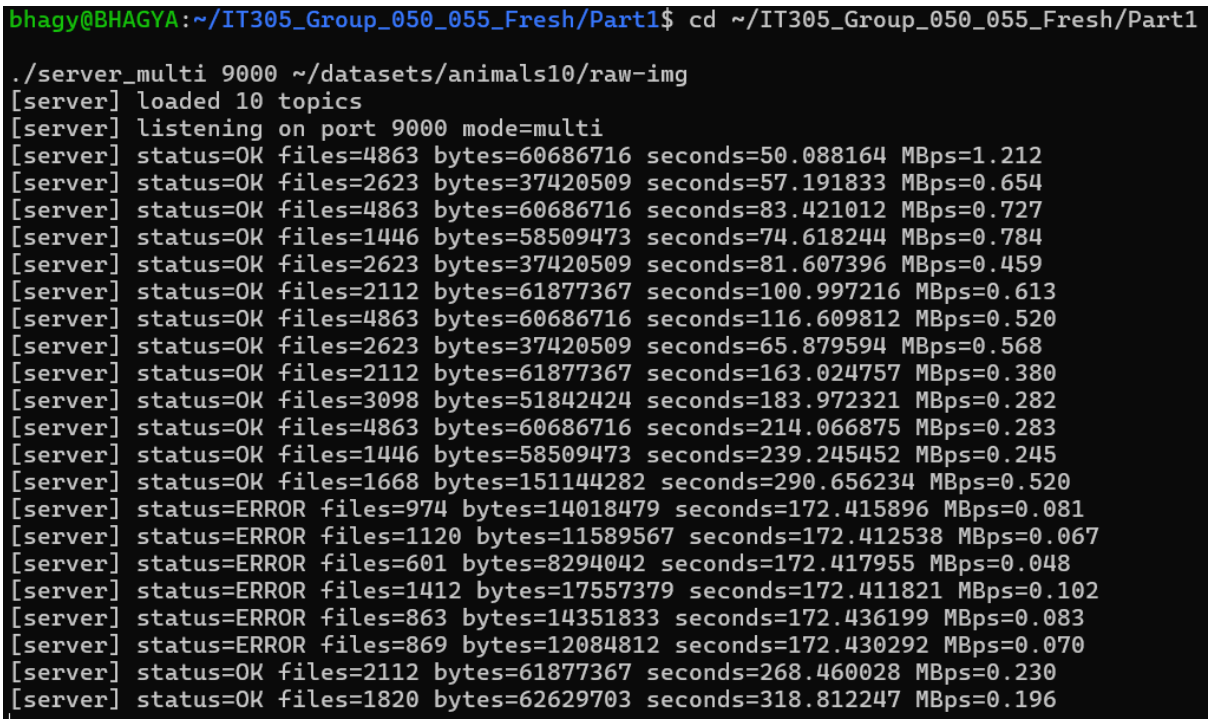
```
========== FINAL MULTI-THREADED CSV ==========
client_count,record_type,client_id,topic_name,transfer_status,files_received,total_bytes_transferred,client_transfer_time_seconds,client_transfer_rate_MBps,batch_wall_time_seconds,aggregate_throughput_MBps
1,client,1,cane,OK,4863,60686716,166.357265,0.365,166.382609,0.0
1,aggregate,0,ALL_TOPICS,OK,0,60686716,0.0,0.0,166.382609,0.364742
2,client,1,cane,OK,4863,60686716,81.975382,0.74,82.004859,0.0
2,client,2,cavallo,OK,2623,37420509,61.234879,0.611,82.004859,0.0
2,aggregate,0,ALL_TOPICS,OK,0,98107225,0.0,0.0,82.004859,1.196359
4,client,1,cane,OK,4863,60686716,113.652754,0.534,113.68769,0.0
4,client,2,cavallo,OK,2623,37420509,81.742508,0.458,113.68769,0.0
4,client,3,elefante,OK,1446,58509473,89.494624,0.654,113.68769,0.0
4,client,4,farfalla,OK,2112,61877367,102.811036,0.602,113.68769,0.0
4,aggregate,0,ALL_TOPICS,OK,0,218494065,0.0,0.0,113.68769,1.92188
6,client,1,cane,OK,4863,60686716,216.444387,0.28,494.636647,0.0
6,client,2,cavallo,OK,2623,37420509,85.223242,0.439,494.636647,0.0
6,client,3,elefante,OK,1446,58509473,448.453088,0.13,494.636647,0.0
6,client,4,farfalla,OK,2112,61877367,181.440421,0.341,494.636647,0.0
6,client,5,gallina,OK,3098,51842424,196.813469,0.263,494.636647,0.0
6,client,6,gatto,OK,1668,151144282,494.491856,0.306,494.636647,0.0
6,aggregate,0,ALL_TOPICS,OK,0,421480771,0.0,0.0,494.636647,0.852102
8,client,1,cane,OK,4863,60686716,32.817372,1.849,63.484055,0.0
8,client,2,cavallo,OK,2623,37420509,16.441789,2.276,63.484055,0.0
8,client,3,elefante,OK,1446,58509473,28.246476,2.071,63.484055,0.0
8,client,4,farfalla,OK,2112,61877367,30.659755,2.018,63.484055,0.0
8,client,5,gallina,OK,3098,51842424,24.560213,2.111,63.484055,0.0
8,client,6,gatto,OK,1668,151144282,63.484055,2.381,63.484055,0.0
8,client,7,mucca,OK,1866,25833171,13.762378,1.877,63.484055,0.0
8,client,8,pecora,OK,1820,62629703,30.194109,2.074,63.484055,0.0
8,aggregate,0,ALL_TOPICS,OK,0,5899436845,0.0,0.0,63.484055,8.032626
```

Terminal Output From Server Side

```
PS C:\Users\bhagy> wsl -d Ubuntu-22.04
bhagy@BHAGYA:/mnt/c/Users/bhagy$ ./server_single 9000 ~/datasets/animals10/raw-img
-bash: ./server_single: No such file or directory
bhagy@BHAGYA:/mnt/c/Users/bhagy$ cd ~/IT305_Group_050_055_Fresh/Part1
bhagy@BHAGYA:~/IT305_Group_050_055_Fresh/Part1$ ./server_single 9000 ~/datasets/animals10/raw-img
[server] loaded 10 topics
[server] listening on port 9000 mode=single
^C[server] stopped
bhagy@BHAGYA:~/IT305_Group_050_055_Fresh/Part1$ ./server_single 9000 ~/datasets/animals10/raw-img
[server] loaded 10 topics
[server] listening on port 9000 mode=single
[server] status=OK files=4863 bytes=60686716 seconds=32.539764 MBps=1.865
[server] status=OK files=4863 bytes=60686716 seconds=28.519013 MBps=2.128
[server] status=OK files=2623 bytes=37420509 seconds=16.570546 MBps=2.258
[server] status=OK files=1446 bytes=58509473 seconds=37.080210 MBps=1.578
[server] status=OK files=2112 bytes=61877367 seconds=43.671249 MBps=1.417
[server] status=OK files=3098 bytes=51842424 seconds=32.751929 MBps=1.583
[server] status=OK files=1668 bytes=151144282 seconds=107.245631 MBps=1.409
[server] status=OK files=1866 bytes=25833171 seconds=13.208420 MBps=1.956
[server] status=OK files=1820 bytes=62629703 seconds=41.717961 MBps=1.501
[server] status=OK files=4821 bytes=78999493 seconds=43.085357 MBps=1.834
[server] status=OK files=1862 bytes=25409193 seconds=6.983396 MBps=3.639
```

```
bhagy@BHAGYA:~/IT305_Group_050_055_Fresh/Part1$ cd ~/IT305_Group_050_055_Fresh/Part1

./server_single 9000 ~/datasets/animals10/raw-img
[server] loaded 10 topics
[server] listening on port 9000 mode=single
[server] status=OK files=4863 bytes=60686716 seconds=30.540217 MBps=1.987
[server] status=OK files=4863 bytes=60686716 seconds=29.037489 MBps=2.090
[server] status=OK files=4863 bytes=60686716 seconds=29.641564 MBps=2.047
[server] status=OK files=2623 bytes=37420509 seconds=19.250577 MBps=1.944
[server] status=OK files=4863 bytes=60686716 seconds=27.726052 MBps=2.189
[server] status=OK files=2623 bytes=37420509 seconds=22.033194 MBps=1.698
[server] status=OK files=2112 bytes=61877367 seconds=32.907801 MBps=1.880
[server] status=OK files=1446 bytes=58509473 seconds=35.973226 MBps=1.626
[server] status=OK files=4863 bytes=60686716 seconds=32.152370 MBps=1.887
[server] status=OK files=2623 bytes=37420509 seconds=24.434755 MBps=1.531
[server] status=OK files=2112 bytes=61877367 seconds=37.182597 MBps=1.664
[server] status=OK files=1446 bytes=58509473 seconds=41.165202 MBps=1.421
[server] status=OK files=3098 bytes=51842424 seconds=27.718425 MBps=1.870
[server] status=OK files=1668 bytes=151144282 seconds=103.761583 MBps=1.457
[server] status=OK files=4863 bytes=60686716 seconds=45.535390 MBps=1.333
[server] status=OK files=1446 bytes=58509473 seconds=38.397607 MBps=1.524
[server] status=OK files=2623 bytes=37420509 seconds=18.233984 MBps=2.052
[server] status=OK files=3098 bytes=51842424 seconds=32.337636 MBps=1.603
[server] status=OK files=2112 bytes=61877367 seconds=34.656854 MBps=1.785
[server] status=OK files=1668 bytes=151144282 seconds=82.677215 MBps=1.828
[server] status=OK files=1866 bytes=25833171 seconds=9.742593 MBps=2.652
[server] status=OK files=1820 bytes=62629703 seconds=45.965367 MBps=1.363
^C[server] stopped
```



Used Command for Connection Check

Test-NetConnection 10.11.20.136 -Port 9000

Other Used Command

mkdir -p ~/IT305_Group_050_055_Fresh
cd ~/IT305_Group_050_055_Fresh/IT305_Group_050_055_Fresh/Part1
./client 10.155.178.136 9000 cane ./downloads